East West University
Course Outline

Department of Mathematical and Physical Sciences

Semester: **Fall 2024**

Course Title: Differential and Integral Calculus **Course Code:** MAT101

Pre-Requisite: None **Credit Hour:** 3

Course Instructor: Kazi Nusrat Islam, Lecturer, Dept. of MPS

Office Room: Room# FUB-1006 **Phone Number:**

E-mail Address: nusrat.islam@ewubd.edu

Class Schedule and Office Hour:

Day	Course Code	Class Time and Room No.	Office Hour
Sunday	MAT102(9)	10:50 am – 12:05 pm FUB - 802	12:10 pm – 01:10 pm 4:00 pm – 4:30 pm
	MAT101(32)	4:30 pm – 5:45 pm FUB - 701	
Monday	MAT102(1)	08:00 am – 09:15 am FUB - 702	10:45 am – 11:45 am 11:45 am – 12:45 pm
	MAT101(20)	9:25 am – 10:40 am AB3 - 602	
Tuesday	MAT101(2)	08:00 am – 09:15 am AB3 - 502	2:30 pm – 3:30 pm 3:30 pm – 4:30 pm
	MAT101(32)	4:30 pm – 5:45 pm FUB - 802	
Wednesday	MAT102(1)	08:00 am – 09:15 am FUB - 702	10:45 am – 11:45 am 11:45 am – 12:45 pm
	MAT101(20)	9:25 am – 10:40 am AB3 - 602	
	MAT101(2)	08:00 am – 09:15 am FUB - 401	9:20 am – 10:50 am
	MAT102(9)	10:50 am – 12:05 pm FUB - 701	

Course Contents

Differential Calculus

Limit, Continuity and Differentiability, Successive differentiation of various types of functions. Leibnitz's theorem, Rolle's theorem, Mean value theorem, Taylor's and Maclaurin's theorems in finite and infinite forms, Lagrange's form of remainders, Cauchy's form of remainders. Expansion of functions. Evaluation of indeterminate forms by L'Hospital rule. Partial differentialtion. Euler's theorem. Tangent and Normal. Concavity of functions. Determination of maximum and minimum values of functions and points of inflection with Applications. Curvature, Asymptotes.

Integral Calculus

Integration by the method of substitution. Standard integrals. Integration by successive reduction. Definite integrals, its properties and use in summing series. Walli's formulae. Improper integrals. Beta function and Gamma function. Area under a plane curve and area of a region enclosed by two cueves in Cartesian and polar co-ordinates. Volumes of solids of revolution. Volume of hollow solids of revolution by shell method. Area of surface of revolution. Jacobians. Multiple integrals with applications.

Course Rationale

The purpose of studying calculus is simply to introduce the mind of students to the scientific method of analysis. Basic skills in calculus help the students to create models to obtain optimal solutions.

Course Objective

The objectives of the course are to

1. understand the basic concepts of differential and integral calculus.
2. learn the usage of mathematical tools to facilitate understanding and visualization of mathematical problems.
3. enable students to apply the acquired knowledge and skills in professional and specialist courses.

Course Learning Outcomes

On completion of this course, the student will be able to

1. define the basic concepts and principles of differential and integral calculus of real functions and series.
2. interpret the geometric meaning of differential and integral calculus.
3. apply the concept and principles of differential and integral calculus to solve physical problems.
4. analyze the properties of functions based on graph obtained using a mathematical tool.

Textbook

Calculus – Howard Anton, Irl Bivens and Stephen Davis, 10 th Edition, John Wiley & Sons, Inc.

Examination Date:

	Section	Date
Mid Term Examination	Section – 20	December 11, 2024
	Section – 32	December 08, 2024
	Section – 2	December 10, 2024
Final Examination	Section – 20	February 12, 2025
	Section – 32	February 9, 2025
	Section – 2	February 11, 2025

Assessment Tools and Mark Distribution:

Mid Term Examination	30%
Final Examination	30%
Class Test	15%
Presentation	5%
Viva Voce	5%
Assignment	5%
Class Performance	5%
Class Attendance	5%
Total	100%

Grading Policy:

Score in 100	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Important Dates:

October 20, 2024	First day of classes
October 29, 2024	Last day to add course(s)
October 29, 2024	Last day to clear Incomplete ("I") grades
October 30, 2024	Last day to drop course(s)/semester with 100% refund
November 04, 2024	Last day of payment of tuition fees without late fee Last day of payment of the 1st Installment: 60% of the tuition fees without late fee
December 10, 2024	Last day of Tuition Payment with Late Fee of Tk. 500/-
December 18, 2024	Last day of Tuition Payment with Late Fee of Tk. 1000/-
January 02, 2025	Last day to drop course(s)/semester with 50% refund
January 09, 2025	Last day to withdrawal course(s)/semester
February 9, 2025 – February 13, 2025	Final Examinations

Special Instruction:

- Students are not allowed to enter into the classroom after 10 minutes of starting time.
- Students should wear proper dress inside the university campus.
- Students are requested to switch off their mobile telephone during class hours.
- No make-up quizzes and assignments will be held.
- Students are requested to go to the Teaching Assistant (TA) regularly.
- There is zero tolerance for cheating at EWU. Students caught with cheat sheets in their possession, whether used or not used, &/or copying from cheat sheets, writings on the palm of hand, back of calculators, chairs, or nearby walls, etc. would be treated as cheating in the exam hall. The only penalty for cheating is expulsion from EWU.

(**Kazi Nusrat Islam**)
Date: 20.10.2024